<u>Set:1</u>

1. Print "Hello, World!“

HelloWorld.java

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

Output

```
Hello, World!

=== Code Execution Successful ===
```

2. Take the user’s name and greet them. Example, Input: “Your Name” and Output: Hello, “Your Name”!

Greeting.java

```java
import java.util.Scanner;

public class Greeting {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your name: ");
        String name = sc.nextLine();

        System.out.println("Hello, " + name + "!");
    }
}
```

Output

```
Enter your name: Romen Paul
Hello, Romen Paul!

=== Code Execution Successful ===
```

3. Check if a number is even or odd

EvenOdd.java

```java
import java.util.Scanner;

public class EvenOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int num = sc.nextInt();

        if (num % 2 == 0)
            System.out.println("Even");
        else
            System.out.println("Odd");
    }
}
```

Output

```
Enter number: 26
Even

=== Code Execution Successful ===
```

4. Find the maximum of 3 numbers: Take 3 numbers from user input, print the largest.

MaxThree.java

```java
import java.util.Scanner;

public class MaxThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        int max = a;

        if (b > max) max = b;
        if (c > max) max = c;

        System.out.println("Max = " + max);
    }
}
```

Output

```
2
4
6
Max = 6

=== Code Execution Successful ===
```

5. Simple Calculator (Add, Subtract, Multiply, Divide): Use if-else or switch to perform perations.

Calculator.java

```java
import java.util.Scanner;

public class Calculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter two numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();

        System.out.print("Enter operator (+ - * /): ");
        char op = sc.next().charAt(0);

        if (op == '+') System.out.println(a + b);
        else if (op == '-') System.out.println(a - b);
        else if (op == '*') System.out.println(a * b);
        else if (op == '/') System.out.println(a / b);
        else System.out.println("Invalid operator");
    }
}
```

Output

```
Enter two numbers: 2 5
Enter operator (+ - * /): *
10

=== Code Execution Successful ===
```

6. Print numbers from 1 to N. Take N from the user input.

OneToN.java

```java

import java.util.Scanner;
public class OneToN {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            System.out.println(i);
        }
    }
}
```

Output

```
5
1
2
3
4
5

=== Code Execution Successful ===
```

7. Print the multiplication table of a number

Table.java

```java
import java.util.Scanner;

public class Table {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for (int i = 1; i <= 10; i++) {
            System.out.println(n + " x " + i + " = " + (n * i));
        }
    }
}
```

Output

```
5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

=== Code Execution Successful ===
```

8. Calculate the factorial of a number. Example: 5! = 5 × 4 × 3 × 2 × 1=120

Factorial.java

```java
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int fact = 1;

        for (int i = 1; i <= n; i++) {
            fact *= i;
        }

        System.out.println("Factorial = " + fact);
    }
}
```

Output

```
5
Factorial = 120

=== Code Execution Successful ===
```

9. Sum of all numbers from 1 to N

SumN.java

```java
import java.util.Scanner;

public class SumN {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int sum = 0;

        for (int i = 1; i <= n; i++) {
            sum += i;
        }

        System.out.println("Sum = " + sum);
    }
}
```

Output

```
5
Sum = 15

=== Code Execution Successful ===
```

Set:2

1. Store and print 5 numbers in an array

```
import java.util.Scanner;

public class ArrayInput {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];

        for (int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }

        for (int i = 0; i < 5; i++) {
            System.out.println(arr[i]);
        }
    }
}
```

2. Find the largest number in an array

```
import java.util.Scanner;

public class ArrayMax {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];

        for (int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }

        int max = arr[0];
```

```
        for (int i = 1; i < 5; i++) {
            if (arr[i] > max) {
                max = arr[i];
            }
        }

        System.out.println("Max = " + max);
    }
}
```

3. Find the smallest number in an array

```
import java.util.Scanner;

public class ArrayMin {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int min = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] < min) {
                min = arr[i];
            }
        }

        System.out.println("Min = " + min);
```

```java
    }
}
```

4. Calculate the average of an array

```java
import java.util.Scanner;

public class ArrayAverage {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        int sum = 0;

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
            sum += arr[i];
        }

        double avg = (double) sum / n;

        System.out.println("Average = " + avg);
    }
}
```

5. Count even and odd numbers in an array

```java
import java.util.Scanner;

public class EvenOddCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
```

```
        int[] arr = new int[n];

        int even = 0, odd = 0;

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();

            if (arr[i] % 2 == 0)
                even++;
            else
                odd++;
        }

        System.out.println("Even = " + even);
        System.out.println("Odd = " + odd);
    }
}
```

6. Search for an element in an array

```
import java.util.Scanner;

public class SearchArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int key = sc.nextInt();
```

```java
        boolean found = false;

        for (int i = 0; i < n; i++) {
            if (arr[i] == key) {
                found = true;
                break;
            }
        }

        if (found)
            System.out.println("Found");
        else
            System.out.println("Not Found");
    }
}
```

7. Sort an array (Ascending Order)

```java
import java.util.Scanner;

public class SortArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - i - 1; j++) {
                if (arr[j] > arr[j + 1]) {
```

```
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }

    for (int i = 0; i < n; i++) {
        System.out.println(arr[i]);
    }
  }
}
```

8. Check if a string is a palindrome

```
import java.util.Scanner;

public class Palindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();
        String rev = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            rev += str.charAt(i);
        }

        if (str.equals(rev))
            System.out.println("Palindrome");
        else
            System.out.println("Not Palindrome");
    }
}
```

9. Count vowels and consonants in a string

```
import java.util.Scanner;

public class VowelConsonant {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine().toLowerCase();
        int v = 0, c = 0;
        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);

            if (ch >= 'a' && ch <= 'z') {
                if ("aeiou".indexOf(ch) != -1)
                    v++;
                else
                    c++;
            }
        }

        System.out.println(v);
        System.out.println(c);
    }
}
```

10. Convert a string to uppercase and lowercase

```
import java.util.Scanner;
public class CaseConvert {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String str = sc.nextLine();
        System.out.println(str.toUpperCase());
        System.out.println(str.toLowerCase());
```

```
    }
}
```

<u>Set:3</u>

1. Find if integer B exists (average of A and C)

Problem1.java

```java
import java.util.Scanner;

public class Problem1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int A = sc.nextInt();
        int C = sc.nextInt();

        if ((A + C) % 2 == 0) {
            System.out.println("Yes, B exists: " + (A + C) / 2);
        } else {
            System.out.println("No such integer B");
        }
    }
}
```

Output

```
10
20
Yes, B exists: 15

=== Code Execution Successful ===
```

2.Word Abbreviation

Problem2.java

```java
import java.util.Scanner;

public class Problem2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String word = sc.next();

        if (word.length() > 10) {
            System.out.println("" + word.charAt(0) + (word.length() - 2) + word
                .charAt(word.length() - 1));
        } else {
            System.out.println(word);
        }
    }
}
```

Output

```
localization
l10n

=== Code Execution Successful ===
```

3.Capitalize first letter

Problem3.java Run

```java
import java.util.Scanner;

public class Problem3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String word = sc.next();

        String result = word.substring(0,1).toUpperCase() + word.substring(1);
        System.out.println(result);
    }
}
```

Output

```
hello
Hello

=== Code Execution Successful ===
```

4.**Find unique number**

Problem4.java Run

```java
import java.util.Scanner;

public class Problem4 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if (a == b) System.out.println(c);
        else if (a == c) System.out.println(b);
        else System.out.println(a);
    }
}
```

Output

```
10 10 15
15

=== Code Execution Successful ===
```

5. Average of A and B > C ?

Problem5.java

Run

```java
import java.util.Scanner;

public class Problem5 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int A = sc.nextInt();
        int B = sc.nextInt();
        int C = sc.nextInt();

        if ((A + B) > 2 * C) {
            System.out.println("Yes");
        } else {
            System.out.println("No");
        }
    }
}
```

Output

```
5 9 6
Yes

=== Code Execution Successful ===
```

6.IIUC-TV subscription cost

Problem6.java

Run

```java
import java.util.Scanner;

public class Problem6 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = sc.nextInt();
        int X = sc.nextInt();

        int subscriptions = (N + 5) / 6;
        System.out.println(subscriptions * X);
    }
}
```

Output

```
10 100
200

=== Code Execution Successful ===
```

7.Assignment completion

Problem7.java

Run

```java
import java.util.Scanner;

public class Problem7 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int X = sc.nextInt();

        if (X + 3 <= 10) {
            System.out.println("Yes");
        } else {
            System.out.println("No");
        }
    }
}
```

Output

```
7
Yes

=== Code Execution Successful ===
```

8.Passed students > 50% ?

Problem8.java

Run

```java
import java.util.Scanner;

public class Problem8 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int X = sc.nextInt();
        int Y = sc.nextInt();
        int Z = sc.nextInt();

        int total = X * Y;

        if (Z > total / 2) {
            System.out.println("Yes");
        } else {
            System.out.println("No");
        }
    }
}
```

Output

```
2 10 12
Yes

=== Code Execution Successful ===
```

9.Total working hours

Problem9.java

Run

```java
import java.util.Scanner;

public class Problem9 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int X = sc.nextInt();
        int Y = sc.nextInt();

        int total = (4 * X) + Y;
        System.out.println(total);
    }
}
```

Output

```
8
4
36

=== Code Execution Successful ===
```

10.Water bottle condition

Problem10.java

Run

Share code

```java
import java.util.Scanner;

public class Problem10 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int B1 = sc.nextInt();
        int B2 = sc.nextInt();
        int B3 = sc.nextInt();

        int empty = (B1 == 0 ? 1 : 0) + (B2 == 0
            ? 1 : 0) + (B3 == 0 ? 1 : 0);

        if (empty >= 2) {
            System.out.println("Water filling
                time");
        } else {
            System.out.println("Not now");
        }
    }
}
```

Output

```
1 0 0
Water filling time

=== Code Execution Successful ===
```

11.Notebook calculation

Problem11.java

Run

```java
import java.util.Scanner;

public class Problem11 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = sc.nextInt();

        int notebooks = (N * 1000) / 100;
        System.out.println(notebooks);
    }
}
```

Output

```
2
20

=== Code Execution Successful ===
```

12.Candy packets

Problem12.java

Run

```java
import java.util.Scanner;

public class Problem12 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = sc.nextInt();
        int X = sc.nextInt();

        int need = Math.max(0, N - X);
        int packets = (need + 3) / 4;

        System.out.println(packets);
    }
}
```

Output

```
10 3
2

=== Code Execution Successful ===
```

13.Contest topic check

Problem13.java

Run

```java
import java.util.Scanner;

public class Problem13 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int A = sc.nextInt();
        int B = sc.nextInt();
        int C = sc.nextInt();
        int X = sc.nextInt();

        if (X == A || X == B || X == C) {
            System.out.println("Yes");
        } else {
            System.out.println("No");
        }
    }
}
```

Output

```
1 3 5 3
Yes

=== Code Execution Successful ===
```

14.Monopoly check

Problem14.java

Run

```java
import java.util.Scanner;

public class Problem14 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int P = sc.nextInt();
        int Q = sc.nextInt();
        int R = sc.nextInt();
        int S = sc.nextInt();

        if (P > Q + R + S || Q > P + R + S || R
            > P + Q + S || S > P + Q + R) {
            System.out.println("Yes");
        } else {
            System.out.println("No");
        }
    }
}
```

Output

```
10 5 3 2
No

=== Code Execution Successful ===
```

15.Temperature agreement

Problem15.java

Run

```java
import java.util.Scanner;

public class Problem15 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int A = sc.nextInt();
        int B = sc.nextInt();
        int C = sc.nextInt();

        if (Math.max(A, C) <= B) {
            System.out.println("Yes");
        } else {
            System.out.println("No");
        }
    }
}
```

Output

```
20 25 22
Yes

=== Code Execution Successful =
```